Curriculum Vitae

Seol Ah Park, Ph.D.

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Faculty of Nuclear Sciences and Physical Engineering,
Czech Technical University in Prague,
Czech Republic

Profile

I am a researcher with expertise in image analysis, mathematical modeling, and computational algorithms, particularly focusing on biomedical data. My primary interests lie in developing image processing and analysis techniques, as well as advanced methods for analyzing and integrating multi-dimensional data.

Research Experience

Postdoctoral Fellow — Mar 2026 – Present
Czech Technical University in Prague, Czech Republic

Research Staff — Oct 2024 – Feb 2026
Slovak University of Technology in Bratislava, Slovakia

Postdoctoral Researcher — Jun 2023 – Oct 2024
Slovak University of Technology in Bratislava, Slovakia

Marie Curie Fellow (Innovative Training Networks) — Sep 2018 – Jun 2021
EU H2020 Marie Skłodowska-Curie Project (ImageInLife, Grant ID: 721537)

Secondments

- LPHI, CNRS, Montpellier, France (Dr. Mai NGUYEN-CHI's team) — Nov 2022 – Dec 2022
- CNRS, Gif-sur-Yvette, France (Dr. Nadine PEYRIERAS' team) — Feb 2020
- Leiden University, Netherlands (Prof. Annemarie MEIJER's team) — Nov 2019 – Dec 2019

Research Assistant

- Institute of Quantum Systems at Chungnam National University — Jul 2015 – Feb 2016
- Korea Advanced Institute of Science and Technology — Apr 2015 – Jun 2015

Education

Ph.D. in Applied Mathematics — Sep 2018 – Jun 2023
Slovak University of Technology in Bratislava, Slovakia
Thesis: "Tracking cells in 2D+time and 3D+time image sequences"

Ph.D. Program in Physics — Mar 2016 – incomplete
Chungnam National University, Republic of Korea
Topic: Modeling neurotransmitter release at the presynapse of an inner hair cell.

M.Sc. in Physics — Mar 2013 – Feb 2015
Chungnam National University, Republic of Korea
Thesis: "Stripe states in honeycomb ribbon using photonic crystals"

B.Sc. in Astronomy & Space Science and Physics (Double Major) — Mar 2009 – Feb 2013
Chungnam National University, Republic of Korea
Theses were replaced by qualified exams.

Supervising Experience

Slovak University of Technology in Bratislava

Bachelor thesis Sep 2025 – Present
"An investigation of dimensionality reduction methods with applications to high-dimensional biomedical data" (internal: Sep 2025 – Feb 2026; external: Mar 2026 – Present)

Bachelor thesis Sep 2024 – Jun 2025
"Exploration of advanced optimization techniques: applications in biology and immunology"

Bachelor thesis Jan 2023 – Jun 2023
"Machine learning methods for cell segmentation in biology data"

Teaching Experience

Slovak University of Technology in Bratislava Feb 2024 – Dec 2025
Informatics 1 & 2, Seminar in Mathematics (Calculus), English Language (Presentation)

Chungnam National University 2016 – 2017
Quantum Mechanics, Electrodynamics

Technical Skills

Programming Languages	Python, C, PyTorch, R, Fortran
Software	Fiji, Paraview, Qtiplot, Origin

Awards

Seal of Excellence, Marie Skłodowska-Curie Postdoctoral Fellowship (PF), score: 95.2/100 2026

2nd KERC Supporter, selected in the Cutting-Edge Biotech category, Korea-EU Research Centre (KERC) 2025

Summa cum Laude, with honors in Astronomy & Space Science, Chungnam National University 2013

Magna cum Laude, with honors in Natural Sciences, Chungnam National University 2013

Received Research Funding

CROP Postdoctoral Fellowship (Czech Technical University in Prague) Mar 2026 – Aug 2027
Marie Skłodowska-Curie COFUND Action

Early Stage Grant STU (Slovak University of Technology), Individual grant (€6,000, conference travel & research expenses) 2024
Project: Phenotypic Classification of Macrophages and Behavioral Analysis

Publications

1. **Park, S.A.**, Lupi, G., Ozbilgic, R., Kollár, M., Özvat, A.A., Nguyen-Chi, M., Mikula, K., "The morpho-kinetic landscape of macrophage modes during wound healing in zebrafish", *Scientific Reports*, **16**, 6506, (2026).

2. Koljančić, N., **Park, S.A.**, Gurnari, D., Dłotko, P., Hahn, J., Špánik, I., "Untargeted Chemical Profiling of Two-Dimensional Gas Chromatography Coupled with High-Resolution Mass Spectrometry Data for Botrytized Wines via Topological Data Analysis", *Analytical Chemistry* **97**, 25853–25867, (2025).

3. Lupi, G., **Park, S.A.**, Ozbilgic, R., Ambroz, M., Lutfalla, G., Nguyen-Chi, M., Mikula, K., "Math-

ematical and numerical methods for understanding immune cell motion during wound healing", *Mathematical Modelling of Natural Phenomena* **20**, 28, (2025).

4. **Park, S.A.** and Mikula, K., "Automated completion of segmented fragments of macrophages using weighted dilation and erosion in 2D+time microscopy videos", *Proceedings of ALGORITMY 2024*, 11–20, (2024).

5. Lupi, G., Mikula, K., **Park, S.A.**, "Macrophages trajectories smoothing by evolving curves", *Tatra Mountains Mathematical Publications*, **87**, 145–170, (2024).

6. **Park, S.A.**, Sipka, T., Kriva, Z., Lutfalla, G., Nguyen-Chi, M., Mikula, K., "Segmentation-based tracking of macrophages in 2D+time microscopy movies inside a living animal", *Computers in Biology and Medicine*, **153**, 106499, (2023).

7. Uba, M.O., Mikula, K., **Park, S.A.**, "4D segmentation algorithm with application to 3D+time image segmentation", *Japan Journal of Industrial and Applied Mathematics*, **40**, 109–139, (2023).

8. Sipka, T., **Park, S.A.**, Ozbilgic, R., Balas, L., Durand, T., Mikula, K., Lutfalla, G., Nguyen-Chi, M., "Macrophages undergo a behavioural switch during wound healing in zebrafish", *Free Radical Biology and Medicine*, **192**, 200–212, (2022).

9. **Park, S.A.**, Sipka, T., Kriva, Z., Ambroz, M., Kollár, M., Kósa, B., Nguyen-Chi, M., Lutfalla, G., Mikula, K., "Macrophage image segmentation by Thresholding and subjective surface Method", *Tatra Mountains Mathematical Publications*, **75**, 103–120, (2020).

10. **Park, S.A.**, Son, Y.W., Ahn, K.H., "Stripe states in photonic honeycomb ribbon", *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, **471**, 20140765, (2015).
    * After this publication, I changed my name from Sul-Ah to Seol Ah to have the same name as my passport.

Submitted

11. **Park, S.A.**, Kim, Y., Gurnari, D., Dłotko, P., Hahn, J., "Spatial analysis of malignant-immune cell interactions in the tumor microenvironment using topological data analysis", *npj Systems Biology and Applications*.

12. Lee, G., **Park, S.A.**, Lee, S., "Data classification framework using phase-field models and its efficient scheme", *Computers & Mathematics with Applications*.

13. Lupi, G., Mikula, K., Nguyen-Chi, M., **Park, S.A.**, "Comparative analysis of macrophage behavior under different wound conditions in zebrafish", *IEEE EMBC 2026*.